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221745

4th Sem / Mechanical Engg.

Subject : Machine Design

Time : 3 Hrs. M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 In machine design, which of the following joins two rotating shafts to each other?

a) Key b) Coupling

c) Gear d) Belt drive

Q.2 The centre to centre distance between two consecutive rivets in a rows is called______

a) Margin b) Pitch

c) Back Pitch d) Diagonal Pitch

Q.3 Resistance to fatigue of a material is measured by

a) Elastic limit b) Young's modulus

c) Bulk modulus d) Endurance limit

Q.4 Maximum shear stress theory is mostly used for

a) Ductile material b) Brittle material

c) Hard material d) Tough material

Q.5 Material used for high strength shaft is

a) Nickel steel

b) Chrome vanadium steel

c) Nickel chromium steel

d) Any of the above

Q.6 In a flange coupling the weekest link should be

a) Flange b) Key

c) Bolt bush d) Shaft

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 Name any two rivet materials.

Q.8 Define fatigue.

Q.9 Draw the symbol of butt joint.

Q.10 The end of rod which is inserted into a socket is known as________

Q.11 Give two applications of springs.

Q.12 What is the material of key.

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Write the advantage and disadvantages of screw joints.

Q.14 Classify the various types of springs with their applications.

Q.15 What do you mean by SN curve? Give its significance in design.

Q.16 Define key. Write the classification of keys.

Q.17 Explain the processes to make the riveted joint leak proof.

Q.18 Describe various types of designs.

Q.19 Write short note on maximum stress theory.

Q.20 Give the name of parts of screw jack with their materials.

Q.21 Draw a screw thread, explaining its terminology.

Q.22 Explain the procedure of design of single rivet lap joint.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 A solid circular shaft is subjected to a bending moment of 3000 N-m and a torque of 10000 N-m. The shaft is made of 45C8 steel having ultimate tensile stress of 700 MPA and an Ultimate shear stress of 500 Mpa. Assuming a factor of safety as 6, determine the diameter of shaft. Consider both of the theories.

Q.24 A rectangular sunk 16mm wide x 12mm thick x 80 mm long key is used to transmit a torque of 25 kN-m from a 100 mm diameter shaft. Calculate the induced shear and crushing stress in the key.

Q.25 Explain the following :

a) Classify various types of joints

b) What is shaft coupling? What are its different types?